

A $p > 2$ obstruction for the restricted isomorphism property into ℓ_r

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Abstract

Friedland and Guedon ask what happens for $p > 2$ in their search for operators satisfying the restricted isomorphism property $\mathcal{P}_1(m)$. Their $0 < p < 2$ construction uses p -stable random variables and yields large sparsity order for operators into ℓ_r^N , $0 < r \leq 1$. We prove that this target family has a simple obstruction when $p > 2$: if $A : \ell_p^n \rightarrow \ell_r^N$, $0 < r \leq 1$, satisfies $\mathcal{P}_1(m)$ with distortion β/α , then

$$m^{1/2-1/p} \lesssim_r \beta/\alpha.$$

Thus fixed-distortion ℓ_r -valued restricted isomorphisms cannot have sparsity order $m \rightarrow \infty$ for $p > 2$. This bound is sharp: normalized Gaussian maps into ℓ_r^N , $N \gtrsim_r m \log(en/m)$, satisfy $\mathcal{P}_1(m)$ with distortion $\lesssim_r m^{1/2-1/p}$. The same lower-bound proof gives an analogous cotype-2 obstruction for Banach targets such as Hilbert spaces and L_1 .

1 Source question

The source defines $\text{sparse}(m) = \{x \in \mathbb{R}^n : |\text{supp } x| \leq m\}$. An operator $A : \ell_p^n \rightarrow E_1$ satisfies property $\mathcal{P}_1(m)$ if there are constants $\alpha, \beta > 0$ such that

$$\forall x \in \text{sparse}(m) \quad \alpha|x|_p \leq \|Ax\|_{E_1} \leq \beta|x|_p.$$

After recalling that many random matrices satisfy this property for $p = 2$, and after announcing a p -stable construction for $0 < p < 2$, the source asks what happens for $p > 2$:

2 The obstruction

Theorem 1. *Let $p > 2$, $0 < r \leq 1$, and $1 \leq m \leq n$. Suppose that*

$$A : \ell_p^n \longrightarrow \ell_r^N$$

satisfies $\mathcal{P}_1(m)$ with constants $\alpha, \beta > 0$, where ℓ_r^N is equipped with the usual quasi-norm $\|z\|_r = (\sum_{j=1}^N |z_j|^r)^{1/r}$. Then

$$m^{1/2-1/p} \leq K_r \frac{\beta}{\alpha},$$

where K_r depends only on r . Consequently, for fixed $p > 2$, r , and fixed distortion β/α , the order m is bounded independently of n and N .

$\text{sparse}(m) = \{x \in \mathbb{R}^n : |\text{supp}(x)| \leq m\}$ the set of vectors in $\mathbb{R}^n$ of cardinality of the support smaller than m . For r -normed quasi-Banach spaces $(E_1, \|\cdot\|_{E_1})$ and $(E_2, \|\cdot\|_{E_2})$ and for $p > 0$, we define two properties of operators from ℓ_p^n into E_j , $j = 1, 2$, which play an important role in this paper.

We say that an operator $A : \ell_p^n \rightarrow E_1$ satisfies property $\mathcal{P}_1(m)$ if

$$\forall x \in \text{sparse}(m) \quad \alpha|x|_p \leq \|Ax\|_{E_1} \leq \beta|x|_p$$

where $|x|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$. This property is a generalization of the Restricted Isometry Property of order δ introduced in [5], for the Euclidean case, that is, $p = 2$, $E_1 = \ell_2$, and the isometry refers to the fact that $\exists \delta \in (0, 1)$ such that $\alpha = 1 - \delta$ and $\beta = 1 + \delta$. We call property $\mathcal{P}_1(m)$ the *restricted isomorphism property*. In the case $p = 2$, some other versions of this property have been considered in the literature [9, 10], introducing the relevance of working with general α, β and also with E_1 being ℓ_1^N instead of a Euclidean space. Here we introduce a general setting that is useful when a quasi-Banach space E_1 has stable type p (see Section 4 for the definition). The main difficulty is to find operators

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²Keywords.

Figure 1: The source definition of $\mathcal{P}_1(m)$.

Proof. Fix a coordinate set $I \subset \{1, \dots, n\}$ with $|I| = m$, and write $y_i = Ae_i \in \ell_r^N$ for $i \in I$. Since each e_i is m -sparse,

$$\|y_i\|_r \geq \alpha, \quad i \in I.$$

Also, for every choice of signs $\varepsilon_i \in \{-1, 1\}$,

$$\left\| \sum_{i \in I} \varepsilon_i y_i \right\|_r \leq \beta \left| \sum_{i \in I} \varepsilon_i e_i \right|_p = \beta m^{1/p}.$$

Let $(\varepsilon_i)_{i \in I}$ be independent Rademacher signs. The lower Khinchine inequality says that for every scalar family $(a_i)_{i \in I}$,

$$\mathbb{E} \left| \sum_{i \in I} \varepsilon_i a_i \right|^r \geq k_r \left(\sum_{i \in I} |a_i|^2 \right)^{r/2}$$

with $k_r > 0$ depending only on r . Applying this coordinatewise gives

$$\begin{aligned} \beta^r m^{r/p} &\geq \mathbb{E} \left\| \sum_{i \in I} \varepsilon_i y_i \right\|_r^r \\ &= \sum_{j=1}^N \mathbb{E} \left| \sum_{i \in I} \varepsilon_i y_i(j) \right|^r \\ &\geq k_r \sum_{j=1}^N \left(\sum_{i \in I} |y_i(j)|^2 \right)^{r/2}. \end{aligned}$$

For $0 < r \leq 1$, and hence $r \leq 2$, the elementary power-mean inequality gives, for each j ,

$$\left(\sum_{i \in I} |y_i(j)|^2 \right)^{r/2} \geq m^{r/2-1} \sum_{i \in I} |y_i(j)|^r.$$

4 Operators satisfying property $\mathcal{P}_1(m)$

In order to apply Theorem 1 we should find operators that satisfy property $\mathcal{P}_1(m)$. For $p = 2$, the set of matrices that satisfy property $\mathcal{P}_1(m)$ is wide. Indeed, random matrices like e.g. Bernoulli matrices, Gaussian matrices, matrices with independent log-concave rows, satisfy this property for a large m with $E_1 = \ell_2^d$ or ℓ_1^d . In the case $p \neq 2$, the situation is more delicate. We shall present a possible answer when $0 < p < 2$, which is based on the notion of p -stable random variables. A natural question consists of asking what happens for $p > 2$. In that case, our method won't work, as the notion of p -stable random variable is not valid anymore.

4.1 Restricted isometry property for quasi-Banach spaces

We need several consequences of well-known results about p -stable random variables. We refer the reader to Chapter 5 of the book [18] and to [22, 2] for the construction of the random operator that we present here. We recall that a real-valued symmetric random variable θ is called p -stable for $p \in (0, 2]$ if its characteristic function is as follows: for some $\sigma > 0$, $\mathbb{E} \exp(it\theta) = \exp(-\sigma|t|^p)$, for any real t . When $\sigma = 1$, we say that θ is standard.

Figure 2: The source paragraph asking about $p > 2$.

Therefore

$$\begin{aligned} \beta^r m^{r/p} &\geq k_r m^{r/2-1} \sum_{i \in I} \sum_{j=1}^N |y_i(j)|^r \\ &= k_r m^{r/2-1} \sum_{i \in I} \|y_i\|_r^r \geq k_r \alpha^r m^{r/2}. \end{aligned}$$

Taking r -th roots yields

$$m^{1/2-1/p} \leq k_r^{-1/r} \frac{\beta}{\alpha}.$$

This is the desired estimate, with $K_r = k_r^{-1/r}$. □

3 Sharpness for ℓ_r -valued maps

Theorem 2. *Let $p > 2$, $0 < r \leq 1$, and $1 \leq m \leq n$. For every $0 < \delta < 1$, if*

$$N \geq C_r \delta^{-2} m \log(en/m),$$

then with positive probability the Gaussian operator

$$Ax = \mu_r^{-1/r} N^{-1/r} (\langle g_1, x \rangle, \dots, \langle g_N, x \rangle), \quad \mu_r = \mathbb{E}|g|^r,$$

where $g_1, \dots, g_N$ are independent standard Gaussian vectors in $\mathbb{R}^n$, satisfies

$$(1 - \delta)|x|_2 \leq \|Ax\|_r \leq (1 + \delta)|x|_2$$

for every m -sparse $x \in \mathbb{R}^n$. Consequently, as an operator $\ell_p^n \rightarrow \ell_r^N$, it satisfies $\mathcal{P}_1(m)$ with

$$\alpha = 1 - \delta, \quad \beta = (1 + \delta)m^{1/2-1/p}.$$

Proof. We use the standard Gaussian restricted-isometry net argument, adapted to the ℓ_r quasi-norm. For a fixed Euclidean unit vector x , the random variables $|\langle g_j, x \rangle|^r$ are independent copies of $|g|^r$. Since $|g|^r$ has sub-exponential tails for $0 < r \leq 1$, Bernstein's inequality gives constants $c_r, C_r > 0$ such that

$$\Pr \left\{ \left| N^{-1} \sum_{j=1}^N |\langle g_j, x \rangle|^r - \mu_r \right| > \varepsilon \mu_r \right\} \leq 2e^{-c_r \varepsilon^2 N}, \quad 0 < \varepsilon < 1.$$

Fix a support $I \subset \{1, \dots, n\}$, $|I| = m$, and take an η -net $\mathcal{N}_I$ of the Euclidean unit sphere in $\mathbb{R}^I$ with $|\mathcal{N}_I| \leq (1 + 2/\eta)^m$. A union bound over all supports and all net points shows that, provided

$$N \geq C_r \varepsilon^{-2} m \log(en/m),$$

we have, with positive probability,

$$(1 - \varepsilon) \mu_r \leq N^{-1} \sum_{j=1}^N |\langle g_j, y \rangle|^r \leq (1 + \varepsilon) \mu_r$$

simultaneously for every $y \in \bigcup_I \mathcal{N}_I$.

On this event, pass from the net to the full sparse Euclidean sphere. For a fixed support I , put

$$M_I = \sup \{ \|Ax\|_r^r : x \in \mathbb{R}^I, \|x\|_2 = 1 \}.$$

If x is in that sphere and $y \in \mathcal{N}_I$ satisfies $\|x - y\|_2 \leq \eta$, the r -triangle inequality gives

$$\|Ax\|_r^r \leq \|Ay\|_r^r + \|A(x - y)\|_r^r \leq (1 + \varepsilon)^r + \eta^r M_I.$$

Thus $M_I \leq (1 + \varepsilon)^r / (1 - \eta^r)$. The reverse estimate follows from

$$\|Ax\|_r^r \geq \|Ay\|_r^r - \|A(x - y)\|_r^r \geq (1 - \varepsilon)^r - \eta^r M_I.$$

Choosing η, ε as sufficiently small constant multiples of δ , with constants depending only on r , yields the stated $(1 \pm \delta)$ estimate after the normalization by $\mu_r^{-1/r} N^{-1/r}$.

Finally, for m -sparse x and $p > 2$,

$$\|x\|_p \leq \|x\|_2 \leq m^{1/2-1/p} \|x\|_p.$$

Combining this with the Euclidean estimate gives $\mathcal{P}_1(m)$ with the displayed constants. $\square$

Corollary 1. *Let $p > 2$, and let E be a Banach space with cotype 2 constant $C_2(E)$. If $A : \ell_p^n \rightarrow E$ satisfies $\mathcal{P}_1(m)$ with constants α, β , then*

$$m^{1/2-1/p} \leq C_2(E) \frac{\beta}{\alpha}.$$

Proof. With $y_i = Ae_i$ as above, the lower part of $\mathcal{P}_1(m)$ gives $(\sum_{i \in I} \|y_i\|^2)^{1/2} \geq \alpha m^{1/2}$. By cotype 2,

$$\alpha m^{1/2} \leq C_2(E) \left(\mathbb{E} \left\| \sum_{i \in I} \varepsilon_i y_i \right\|^2 \right)^{1/2}.$$

The upper part of $\mathcal{P}_1(m)$, applied to all signed sums, bounds the last factor by $\beta m^{1/p}$. $\square$

4 Interpretation

The source's $0 < p < 2$ construction obtains m proportional to $\eta n / \log(1 + 1/\eta)$ for operators into ℓ_r^m , $0 < r \leq 1$. Theorem 1 shows that no analogous result can hold for $p > 2$ with the same target family and fixed distortion. This is independent of the random model: it excludes every linear operator $A : \ell_p^m \rightarrow \ell_r^N$. Theorem 2 shows that the obstruction is the right one: the best possible $p > 2$ behavior for ℓ_r -targets is the Euclidean sparse behavior, with unavoidable distortion of order $m^{1/2-1/p}$.

Corollary 1 also explains why the Hilbert and ℓ_1^N targets mentioned in the $p = 2$ discussion cannot support a large-order $\mathcal{P}_1(m)$ theory for $p > 2$ with uniform constants. Hilbert spaces and L_1 -spaces have cotype 2 with universal constants.

This is a partial answer only. It does not rule out all possible quasi-Banach targets for $p > 2$; it rules out the paper's natural ℓ_r -target extension with fixed distortion, gives the sharp distortion tradeoff for ℓ_r -targets, and rules out the standard Banach targets of cotype 2 when one asks for large m with uniform constants.

5 Novelty check

The run indexes were searched for arXiv:1107.0992, the title, restricted isomorphism property, $p > 2$, sparsity, and related embedding keywords. No prior packet was found. Web searches on 2026-06-19 for the exact title with “ $p > 2$ ”, for “property $\mathcal{P}_1(m)$ $p > 2$ ”, and for restricted isomorphism property of ℓ_p into ℓ_r found the source paper but no later exact answer to this $p > 2$ gap.

References

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